The Ω Ledger

One Number, Every Channel: the Consolidated Falsifiability Statement of Finite Ring Cosmology

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Abstract

The FRC programme has one free quantity, and the corpus has already spent it: the cardinality Ω of the finite totality, with derived scales $\sqrt{\Omega}$ (the locality and coherence horizon, the Planck mass in cell units) and $\Omega^{1/4}$ (the observer decidability horizon). The gravity companion reads $\sqrt{\Omega} \sim 10^{61}$ off the rotation-curve acceleration floor, fixing $\Omega \sim 10^{122}$. This note consolidates the entire experimental-contact programme — notes I through V, ten validation suites, the companion manuscript — into a single table in which every channel constrains that one number: three channels impose floors (the horizon must lie beyond the scale already probed), six are exact Ω-independent nulls (falsifiable at every carrier scale, with no parameter to absorb a failure), one is a filed forward prediction, and one is the anchor. The joint consistency window $[8 \times 10^{49}, \infty)$ contains the anchor with seventy-two orders of magnitude of margin, and the binding floor is supplied not by a physics experiment but by number theory: the verified height of the Riemann zeros, through the decidability horizon $\Omega^{1/4}$. The ledger is the programme's standing falsifiability statement: every future improvement of any bound re-tests the same single number, any floor crossing the anchor kills the framework, and the framework cannot retreat, because Ω is already committed elsewhere. Machine-checkable form: `omega.py` (four checks, all passing).

1 The one number

A theory's falsifiability is measured by what it is *not allowed* to do when an experiment disagrees. Collapse models retune their rates; modified gravities retune their potentials; the FRC programme has nothing to retune. Its single scale Ω enters every sector at once — the gravity companion's locality horizon $m_P = \sqrt{\Omega}$ and acceleration floor [2], the Riemann realisation's decidability window $\Omega^{1/4}$ and coherence horizon $\sqrt{\Omega}$ [3], the quantum companion's Born readout window [1], the wrap thresholds of the Sorkin analysis [4], and the granularity ceiling [8] — so a measurement in any one sector is a constraint on all of them. The corpus's working value, read from the rotation-curve floor and consistent with the holographic degree-of-freedom count, is

$$\sqrt{\Omega} \;\sim\; 10^{61}, \qquad \Omega \;\sim\; 10^{122}, \qquad \Omega^{1/4} \;\sim\; 10^{30.5}. \tag{1}$$

2 The ledger

Table 1: The Ω ledger, June 2026. Types: *anchor* fixes Ω; *floor* is a lower bound from a null result at the scale probed; *indep* is an exact null at every carrier scale; *forward* is a filed prediction awaiting data.

channel	type	constraint	basis and falsification trigger
SPARC acceleration floor [2]	anchor	$\sqrt{\Omega} \sim 10^{61}$	rotation-curve floor; fails if the floor is environment-dependent
Riemann zero verification [3]	floor	$\Omega > 8 \times 10^{49}$	zeros on-line to height $3 \times 10^{12} < \Omega^{1/4}$; *binding floor*; an off-line zero below $\Omega^{1/4}$ falsifies
Sorkin nullity [4]	floor	$\Omega > 10^{22}$	$\kappa = 0$ at the largest coherent tallies; any confirmed $\kappa \neq 0$ falsifies at every scale
Born readout window [1]	floor	$\Omega > 10^{22}$	laboratory tallies sub-horizon; quantised Born anomalies would falsify
Granularity ceiling [8]	floor	$\Omega > 10^{18}$	no Born anomaly at achieved coherent depths
GRB dispersion [4]	indep	exact null	transport identities exact; detection of vacuum dispersion falsifies
CHSH/Tsirelson [5]	indep	$S = 2\sqrt{2}$ exact	loophole-free Bell agreement; $S \neq 2\sqrt{2}$ at saturating settings falsifies
Network game [6]	indep	complex table exact	real-vs-complex experiments decide for the quarter-turn core
No-collapse [7]	indep	exact null	1.7×10^5 amu interference passed; confirmed spontaneous collapse falsifies
Equivalence principle [7]	indep	$\eta = 0$ exact	MICROSCOPE 10^{-15} consistent; any EP violation falsifies
BMV rate [6]	forward	corrections $\sim 10^{-61}$	entanglement at the Newtonian rate, $V^2 + C^2 = 1$; null result falsifies the gravitational sector

Proposition 2.1 (Joint consistency)**.** *Every floor lies below the anchor, with margins of* 72, 100, 100, *and* 104 *orders of magnitude; the joint window* $[8 \times 10^{49}, \infty)$ *contains* $\Omega = 10^{122}$*; and the corpus scale usages* $\sqrt{\Omega}$, $\Omega^{1/4}$ *are coherent powers of the one* Ω. Machine-verified (`omega.py`, O1–O3).

Remark 2.2 (The binding floor is arithmetic)**.** The strongest current constraint on the carrier scale comes from no physics laboratory: it is the verified height of the Riemann zeros, raised to the fourth power through the decidability horizon. Number theory presently polices the FRC carrier more tightly than any interferometer — a fitting circumstance for a programme in which the Riemann spectrum is the energy spectrum of the totality's clock [1, 3] — and large-scale zero verification, which continues independently of physics, pushes the binding floor upward with every extension. The floor reaches the anchor only at verification height $\sim 10^{30}$, far beyond computational reach; but *any single off-line zero* found below $\Omega^{1/4}$ falsifies the framework outright, at any time, from a desk.

Remark 2.3 (How the ledger falsifies)**.** Three distinct failure modes, none absorbable. (i) *A floor crosses the anchor*: some null channel probes past 10^{122} and still sees nothing where FRC requires wrap quantisation, or sees something where FRC requires exactness — the single Ω cannot move, because the rotation-curve anchor has already spent it. (ii) *An exact null fails*: $\kappa \neq 0$ at any precision, vacuum dispersion, spontaneous collapse, an EP violation, or sub-Tsirelson saturation — each is forbidden at *every* carrier scale, so no value of Ω rescues the framework. (iii) *The forward prediction misses*: BMV-class experiments find no entanglement at the Newtonian rate, or find a non-Newtonian rate, or violate the $V^2 + C^2 = 1$ complementarity. The framework's distinctive epistemic posture — one committed number, many exact nulls, one dated prediction — is the strongest falsifiable position a finished formalism can occupy, and the ledger is its standing public statement.

3 Standing protocol

The ledger is to be kept current: each improvement in zero-verification height, coherent tally records, interference mass records, EP bounds, dispersion bounds, Bell-test precision, or the arrival of BMV data updates one row and re-tests the same Ω. The machine-checkable form (`omega.py`) encodes every row as an inequality or an exactness flag; a future bound that breaks joint satisfiability will fail check O1 or O2 mechanically. The programme's experimental tier list T1–T9 is closed; the ledger is what remains permanently open — by design, since it is where the framework lives exposed.

Reproducibility

The script `omega.py` resides in the `validation/` folder of the companion draft, slated for https://github.com/gamayos/frc-quantum-numerics. Recorded output: four `PASS` lines, the printed ledger, and `omega: all checks passed`. Programme totals across notes I–V and this consolidation: eleven validation suites, seventy exact checks, all passing.

References

[1] Y. Akhtman. *Quantum Observation in Finite Ring Cosmology.* Companion draft, June 2026.

[2] Y. Akhtman and E. Voether. *Gravitation as Phase Synchronisation on a Finite Substrate.* Preprint, 2026.

[3] Y. Akhtman. *Finite Field Realisation of the Riemann Hypothesis.* Preprint, 2026.

[4] Y. Akhtman. *Experimental Contact I: Exact Nullity Tests.* Companion note, June 2026.

[5] Y. Akhtman. *Experimental Contact II: The Composite Gate.* Companion note, June 2026.

[6] Y. Akhtman. *Experimental Contact III: The Network Game and the Gravitational Channel.* Companion note, June 2026.

[7] Y. Akhtman. *Experimental Contact IV: The Decoherence Floor and the Equivalence Principle.* Companion note, June 2026.

[8] Y. Akhtman. *Experimental Contact V: The Granularity Ceiling and the Emulation Programme.* Companion note, June 2026.